

Linear response of molecular polaritons

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In this article, we show that the collective light-matter strong coupling regime, where N molecular emitters couple to the photon mode of an optical cavity can be mapped to a quantum impurity model where the photon is the impurity that is coupled to a bath of anharmonic transitions. In the thermodynamic limit where $N \gg 1$, we argue that the bath can be replaced with an effective harmonic bath, leading to a dramatic simplification of the problem into one of coupled harmonic oscillators. We derive simple analytical expressions for linear optical spectra (transmission, reflection, and absorption) where the only molecular input required is the molecular linear susceptibility. This formalism is applied to a series of illustrative examples showcasing the role of temperature, disorder, vibronic coupling, and optical saturation of the molecular ensemble, explaining that it is useful even when describing an important class of nonlinear optical experiments. For completeness, we provide a comprehensive Appendix that includes a self-contained derivation of the relevant spectroscopic observables for arbitrary anharmonic systems (for both large and small N) within the rotating-wave approximation. While some of the presented results herein have already been reported in the literature, we provide a unified presentation of the results as well as new interpretations that connect powerful concepts in open quantum systems and linear response theory with molecular polaritonics.

I. INTRODUCTION

Polaritons are hybrid light-matter modes that emerge when the rate of coherent energy exchange between material transitions and a confined electromagnetic mode surpasses their respective dephasing rates. Hereafter, we shall be concerned with molecular polaritons, where the material transitions involve electronic and rovibrational states of molecules. This light-matter strong coupling (SC) regime pertains to the interaction of a large ensemble of $N \gg 1$ molecules with a cavity mode [1]. The latter requirement is a consequence of light-matter coupling per molecule being negligible in standard Fabry-Perot microcavities, thus necessitating a mesoscopic ensemble of transitions to collectively give rise to SC. Such a restriction is lifted in special nanophotonic environments, where SC with small N has been achieved in various platforms [2–6]. Throughout this article we shall be concerned with the large N case, unless otherwise stated.

Over the last decade, molecular microcavities have drawn attention for their potential impact on altering rates and selectivity of chemical reactions [7, 8], enhancing energy transfer processes [9–11], and enabling room-temperature polariton condensation [12, 13], among other exciting phenomena. Linear spectroscopy as in the measurement of linear transmission, absorption, and reflection, has been widely used to characterize many of these experiments and to demonstrate the onset of the SC regime [14]. While a number of experimental studies use classical optics (transfer matrix methods [15–18]) to successfully model their spectra [19–22], a variety of quantum optical methods based on input-output theory [23–26] have also been used to simulate molecular polaritons featuring a simplified energy level structure [27–31]. The latter approach seems to be at odds with the

standard philosophy of computational molecular spectroscopy, where the linear spectrum of a complex molecular system is obtained through the time-dependent calculation of a dipole-dipole correlation function [32–35]. In this article, we show that the calculation of a photon-photon correlation function is enough to obtain all linear optical spectra of the molecular microcavity. Moreover, we prove that in the $N \rightarrow \infty$ limit, these spectra simplify dramatically and can be computed by direct input of the molecular linear susceptibility (which can be obtained by Fourier transformation of the dipole-dipole correlation function), as shown in [36, 37], without running an explicit calculation of N molecules coupled to a photon mode. While these expressions have already been featured in works by Keeling and coworkers [36, 38], we clarify their assumptions and applicability, and provide a new and enlightening interpretation by regarding polaritons as quantum impurity models [39].

This article is structured in the following way. Section II provides the setup of the polariton Hamiltonian and shows how it can be regarded as a quantum impurity model where the photon mode as the impurity (the system) is coupled to a large bath of anharmonic molecular transitions. In the limit of $N \rightarrow \infty$, the reduced dynamics of the photon can be obtained by replacing this complex bath with a surrogate one of harmonic modes, rendering the problem to one of coupled harmonic oscillators. The spectral density of this surrogate bath turns out to be proportional to the imaginary part of the linear susceptibility of the bare molecular ensemble, $J^{\text{eff}}(\omega) \propto \Im \chi(\omega)$. Importantly, this $\chi(\omega)$ does not need to obey Boltzmann statistics for a given temperature, but can correspond to a nonequilibrium stationary state, rendering our formalism applicable even to nonlinear optical experiments where the latter are optically prepared. Section III outlines the expressions for the linear spectra for arbitrary N in terms of the photon Green's function and, for $N \rightarrow \infty$ shows its explicit relation to $\chi(\omega)$. Sec-

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tion IV provides several illustrative examples that are of interest to current polaritonics experiments, including effects of optical saturation, disorder, vibronic coupling, and nonequilibrium stationary states, which can all be treated with the same formalism. Finally, the Appendix provides a self-contained derivation of (A) input-output theory, (B) a derivation of spectroscopic observables based on Kubo linear response and the expression of transmission in the form of a “Landauer formula”, and (C) the simplification of spectra in the harmonic limit ($N \rightarrow \infty$) using Heisenberg equations of motion or, alternatively, using Kubo linear response.

For a practical use of the results in the article, we recommend the reader to directly refer to Table I, which shows the direct relation between the polariton spectra and $\chi(\omega)$ when $N \rightarrow \infty$, and to work through some of the examples in Section IV. Readers who are interested in the theoretical connections between quantum impurity models, open quantum systems, and polaritons are welcome to read Section II. The Appendix can be consulted to understand the derivations of the spectroscopic formulas.

II. MOLECULAR POLARITONS AS QUANTUM IMPURITY PROBLEMS

In the collective SC regime, we are interested in a (harmonic) cavity mode of frequency ω_{ph} coupled to N non-interacting quantum emitters. The Hamiltonian describing this setup (hereafter denoted as the “molecular microcavity”) is

$$H = H_0 + V, \quad (1)$$

where

$$H_0 = H_{ph} + H_{mol} \quad (2)$$

is the zeroth order contribution describing the photon and molecular degrees of freedom,

$$H_{ph} = \hbar\omega_{ph}a^\dagger a, \quad (3)$$

$$\begin{aligned} H_{mol} &= \sum_i H_i(\mathbf{q}_i, \mathbf{Q}_i) \\ &= \sum_y \hbar\Omega_y |y\rangle\langle y|. \end{aligned} \quad (4)$$

Here $[a, a^\dagger] = 1$ and the molecular term is a sum of contributions of different molecules i , each of which depends on its respective electronic and nuclear degrees of freedom $\mathbf{q}_i, \mathbf{Q}_i$. We formally decompose H_{mol} as a sum over many-body eigenstates $|y\rangle$. Finally, for concreteness, the light-matter interaction is taken to be dipolar,

$$V = -\hbar\lambda(a + a^\dagger)\mu, \quad (5)$$

where $\mu = \sum_i \mu_i(\mathbf{q}_i, \mathbf{Q}_i)$ is the dipole operator and $\hbar\lambda = \sqrt{\frac{\hbar\omega_{ph}}{2\epsilon_0\mathcal{V}}}$ is the vacuum electric field (ϵ_0 is the permittivity of vacuum and $\mathcal{V}$ is the cavity mode volume). Here, notice that the “counterrotating” light-matter interaction terms are present; the diamagnetic terms that are proportional to $(a + a^\dagger)^2$ can always be removed via a Bogoliubov transformation [36].

At this point, it is conceptually convenient to invoke nomenclature from the open quantum systems literature and regard the photon mode as the *system* and the molecular degrees of freedom as the *bath*. This scenario corresponds to the archetypal problem of a quantum impurity coupled to a large environment (the standard collective strong-coupling scenario consists of a large number of emitters $N \approx 10^3 - 10^{10}$). Makri [39] has rigorously shown that if the environment is constituted by an infinite number of independent degrees of freedom, and if the system and the bath start in an uncorrelated state,

$$\begin{aligned} \rho(t_{\text{in}}) &= \rho_{ph} \otimes \rho_{mol} \\ &= \rho_{ph} \otimes \sum_y p_y |y\rangle\langle y|, \end{aligned} \quad (6)$$

where we take ρ_{mol} to be an arbitrary stationary density matrix with respect to H_{mol} , with p_y being the probability of being in eigenstate $|y\rangle$ ([39] discusses only the thermal case; the extension to arbitrary stationary states, which is relevant in molecular polaritonics, is presented below). The *reduced system dynamics* can be obtained exactly using a surrogate quantum impurity Hamiltonian H^{eff} ,

$$H^{\text{eff}} = H_0^{\text{eff}} + V^{\text{eff}}, \quad (7)$$

where we replace the original (in general, anharmonic) bath with a surrogate harmonic bath,

$$H_{mol}^{\text{eff}} = \sum_j \hbar\omega_j b_j^\dagger b_j, \quad (8)$$

the system remains the same, $H_{ph}^{\text{eff}} = H_{ph}$,

$$H_0^{\text{eff}} = H_{mol}^{\text{eff}} + H_{ph}, \quad (9)$$

and the interaction between the system and the bath is,

$$\begin{aligned} V^{\text{eff}} &= -(a + a^\dagger) \sum_j c_j x_j \\ &= -(a + a^\dagger) \sum_j \hbar\bar{c}_j (b_j + b_j^\dagger). \end{aligned} \quad (10)$$

The couplings $\hbar\bar{c}_j = c_j \sqrt{\frac{\hbar}{2m_j\omega_j}}$ are characterized by an effective spectral density

$$J^{\text{eff}}(\omega) = \Theta(\omega)\pi\hbar \sum_j |\bar{c}_j|^2 \delta(\omega - \omega_j). \quad (11)$$

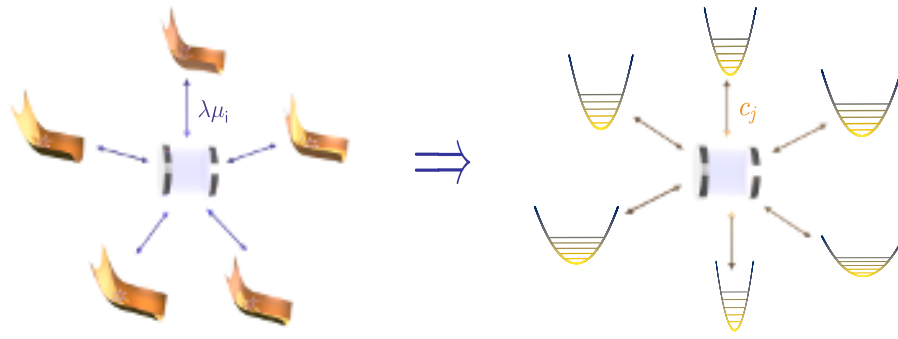


Figure 1. The molecular polariton problem, where the anharmonic degrees of freedom of a large number N of molecules are coupled to a single (harmonic) photon mode can be regarded as a quantum impurity model. When $N \rightarrow \infty$, the reduced dynamics of the photon can be computed exactly by replacing the molecular degrees of freedom with a surrogate harmonic bath.

Here, we adopt the convention that the spectral density is only finite for positive frequencies [the step function $\Theta(\omega) = 1$ for $\omega \geq 0$ and $\Theta(\omega) = 0$ otherwise]. This spectral density is chosen so that the two-point correlation function of the effective bath,

$$C_2^{\text{eff}}(t) = \sum_j |c_j|^2 \langle e^{iH_{\text{mol}}^{\text{eff}}t/\hbar} x_j e^{-iH_{\text{mol}}^{\text{eff}}t/\hbar} x_j \rangle \quad (12)$$

coincides with that of the original system,

$$C_2^{\text{eff}}(t) = C_2(t) \quad (13)$$

for all $t \geq 0$, where

$$\begin{aligned} C_2(t) &= |\hbar\lambda|^2 \langle e^{iH_{\text{mol}}t/\hbar} \mu e^{-iH_{\text{mol}}t/\hbar} \mu \rangle \\ &= |\hbar\lambda|^2 \sum_i \langle e^{iH_{\text{mol}}t/\hbar} \mu_i(\mathbf{q}_i, \mathbf{Q}_i) e^{-iH_{\text{mol}}t/\hbar} \mu_i(\mathbf{q}_i, \mathbf{Q}_i) \rangle. \end{aligned} \quad (14)$$

The trace in Eq. 14 is carried out with respect to ρ_{mol} .

Let us now be more explicit about the evaluation of Eq. 12. Its trace is carried out with respect to an initial stationary state of harmonic oscillators with a *frequency-dependent* inverse temperature $\beta^{\text{eff}}(\omega)$. This complexity arises from the arbitrariness of ρ_{mol} ; in the special case when ρ_{mol} corresponds to a thermal state at inverse temperature $\beta = (k_B \mathcal{T})^{-1}$, the surrogate setup is also thermal at effective inverse temperature $\beta^{\text{eff}}(\omega) = \beta$ [40]. This initial state does not need to be explicitly specified, given the well-known result connecting the two-point correlation function and the spectral density for a bath of harmonic oscillators [41],

$$\begin{aligned} C_2^{\text{eff}}(t) &= \frac{\hbar}{\pi} \int_{-\infty}^{\infty} d\omega' J^{\text{eff}}(\omega') \\ &\times \left[\coth \left(\frac{\beta^{\text{eff}}(\omega') \hbar \omega'}{2} \right) \cos(\omega' t) - i \sin(\omega' t) \right]. \end{aligned} \quad (15)$$

Computing its Fourier transforms (see convention in Eq. 23) at $\pm\omega$,

$$\begin{aligned} C_2^{\text{eff}}(\omega) &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} C_2^{\text{eff}}(t) \\ &= -i\hbar J^{\text{eff}}(\omega) \left[\coth \left(\frac{\beta^{\text{eff}}(\omega) \hbar \omega}{2} \right) + 1 \right], \end{aligned} \quad (16a)$$

$$\begin{aligned} C_2^{\text{eff}}(-\omega) &= -i \int_{-\infty}^{\infty} dt e^{-i\omega t} C_2^{\text{eff}}(t) \\ &= -i\hbar J^{\text{eff}}(\omega) \left[\coth \left(\frac{\beta^{\text{eff}}(\omega) \hbar \omega}{2} \right) - 1 \right], \end{aligned} \quad (16b)$$

and using Eq. 13, we obtain

$$\beta^{\text{eff}}(\omega) = \frac{1}{\hbar\omega} \ln \frac{C_2(\omega)}{C_2(-\omega)}. \quad (17)$$

This expression is essentially a frequency-dependent detailed balance relationship at inverse temperature $\beta^{\text{eff}}(\omega)$. This is a consequence of the fact that linear response of a n -level anharmonic system can be interpreted as a sum of pairwise transitions, each having its own $\beta^{\text{eff}}(\omega)$ [33]. With $C_2^{\text{eff}}(t)$ and $\beta^{\text{eff}}(\omega)$ in hand, Eq. 15 can be solved for $J^{\text{eff}}(\omega)$,

$$\begin{aligned} J^{\text{eff}}(\omega) &= \frac{2}{\hbar} \tanh \frac{\hbar\omega\beta^{\text{eff}}(\omega)}{2} \int_0^{\infty} \Re C_2^{\text{eff}}(t) \cos \omega t dt \\ &= \frac{i\Theta(\omega)}{\hbar} \int_{-\infty}^{\infty} C_2(t) \sin \omega t dt. \end{aligned} \quad (18)$$

Thus, knowledge of $C_2(t)$ of the original bath (Eq. 14) together with the expression in Eq. 18 give rise to $J^{\text{eff}}(\omega)$ which, we shall emphasize, varies as a function of ρ_{mol} . It is well understood that knowledge of the spectral density alone completely characterizes the reduced system dynamics if the bath is composed of independent harmonic oscillators [41].

The intuition behind this mapping is the following. In general, the reduced system dynamics depends on n -point correlation functions of the bath ($n \geq 2$). However, Makri has shown that when the bath is large enough ($N \gg 1$), only the two-point correlation function $C_2(t)$ becomes relevant (the $n > 2$ such functions decay as $O(N^{-1/2})$ if $c_i = O(N^{-1/2})$, which is what happens in molecular polariton experiments, as $c_i \sim O[(\text{Rabispitting})/N^{1/2}]$, where the Rabi splitting is an experimental energy scale that remains finite). It is also well known that a system coupled to a harmonic bath linearly through each of its coordinates x_j (see Eq. 10) has vanishing such functions for $n > 2$. This result is essentially a consequence of the central limit theorem. Hence, we are entitled to replace the original bath with a surrogate harmonic bath so long as the two-point correlation functions coincide. This mapping is useful given that a plethora of tools to solve for the reduced dynamics of systems coupled to harmonic baths have been developed over the last decades.

Note that Eq. 14 is just the usual dipole-dipole correlation function, which can be re-expressed in terms of eigenstates of H_{mol} ,

$$C_2(t) = \lim_{\gamma \rightarrow 0^+} \sum_{y,z} \{p_y \times |\hbar \lambda \langle z | \mu | y \rangle|^2 e^{-i(\omega_{zy} - i\frac{\gamma}{2})t}\}, \quad (19)$$

where $\omega_{zy} = \frac{E_z - E_y}{\hbar}$. Inserting Eq. 19 into Eq. 18 gives rise to

$$\begin{aligned} J^{\text{eff}}(\omega) &= \Theta(\omega) \pi \hbar \sum_{y,z} (p_y - p_z) |\lambda \langle z | \mu | y \rangle|^2 \delta(\omega_{zy} - \omega) \\ &= \Theta(\omega) \hbar \Im \chi(\omega), \end{aligned} \quad (20)$$

where

$$\chi(\omega) = - \lim_{\gamma \rightarrow 0^+} \sum_j \frac{|\bar{c}_j|^2}{\omega - \omega_j + i\frac{\gamma}{2}} \quad (21a)$$

$$= - \lim_{\gamma \rightarrow 0^+} \sum_{y,z} [p_y - p_z] \frac{|\lambda \langle z | \mu | y \rangle|^2}{\omega - \omega_{zy} + i\frac{\gamma}{2}} \quad (21b)$$

Here, $\chi(\omega)$ is the molecular linear susceptibility [33] generalized to arbitrary initial stationary states, where we emphasize that $|y\rangle$ and $|z\rangle$ are eigenstates of the entire molecular ensemble. Thus, we have reached the physically appealing conclusion that the effective spectral density of the bath that couples to the photon mode is simply the absorption spectrum $[\Im \chi(\omega)]$ of the original molecular ensemble. Note that the damping coefficient $\gamma \rightarrow 0$ is introduced to ensure causality in the molecular response (Eq. 19). All molecular line-broadening effects in the spectra comes from the discrete sum in the $\chi(\omega)$ in Eq. 21b [42]. As we shall see below, for our purposes of modelling linear spectra of polaritons, it is more convenient

to work with $\chi(\omega)$ directly than with $J^{\text{eff}}(\beta, \omega)$. However, the former can be readily obtained from the latter; comparing Eqs. 11 and 21, we have that for $\omega \geq 0$,

$$\chi(\omega) = - \lim_{\gamma \rightarrow 0^+} \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{J^{\text{eff}}(\omega')}{\omega - \omega' + i\frac{\gamma}{2}}, \quad (22)$$

while for $\omega < 0$ we can use $\chi(-\omega) = \Re \chi(\omega) - i \Im \chi(\omega)$. Alternatively, $\chi(\omega)$ can be obtained from a Fourier transform of the dipole correlation function in Eq. 14 as [33],

$$\chi(\omega) = - \frac{1}{\hbar^2} \lim_{\gamma \rightarrow 0^+} [C(\omega) + C^*(-\omega)]$$

where we use the following Fourier transform convention (the $-i$ is added to be consistent with a standard convention of Green's function theory, which will be invoked later),

$$f(\omega) = -i \int_{-\infty}^{\infty} dt e^{i\omega t} f(t), \quad (23a)$$

$$f(t) = \frac{i}{2\pi} \int_{-\infty}^{\infty} d\omega e^{-i\omega t} f(\omega). \quad (23b)$$

III. LINEAR SPECTROSCOPY

A. Formulas for arbitrary N

The spectroscopic signals of cavity polaritons can be rigorously obtained via input-output theory [23–26]. For simplicity, we provide expressions under the rotating-wave approximation (RWA) for light-matter interaction (see Eq. 5), thus postponing discussion of ultrastrong coupling systems for another study (at the level of the RWA, the physics is already rich enough),

$$V = -\lambda a \mu^> - \lambda a^\dagger \mu^<, \quad (24)$$

where $\mu^> = \sum_{\omega_{zy} > 0} \langle z | \mu | y \rangle |z\rangle \langle y|$ is the dipole operator projected onto uphill transitions, and $\mu^< = \sum_{\omega_{zy} < 0} \langle z | \mu | y \rangle |z\rangle \langle y|$ is the corresponding projector onto downhill transitions [43].

The central quantity to compute is the photon retarded Green function,

$$D^R(\omega) = -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) \langle e^{iHt/\hbar} a e^{-iH't/\hbar} a^\dagger \rangle, \quad (25)$$

where

$$H' = H - i \frac{\hbar \kappa}{2} a^\dagger a \quad (26)$$

is the effective non-Hermitian Hamiltonian which accounts for coupling of the cavity photon with the left

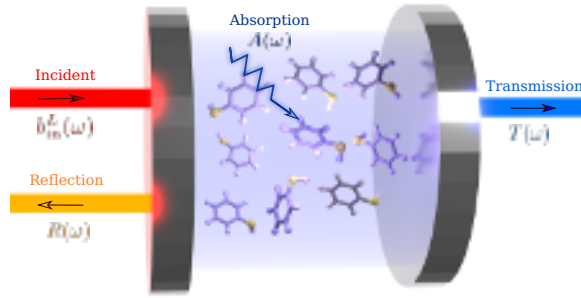


Figure 2. Linear spectroscopy of molecular polaritons as transmission, absorption, and reflection (see Eq. 27).

and right photon continua, with $\kappa = \kappa_L + \kappa_R$, and κ_L and κ_R labeling the corresponding rates of photon escape (hereafter, we use the notation $X' = X - i\frac{\hbar\kappa}{2}a^\dagger a$ for any operator X). Note that the first “ H ” in Eq. 25 is not primed while the second one is (see explanation around Eq. B11). Eq. 25 yields frequency resolved transmission, reflection, and absorption spectra,

$$T(\omega) = \kappa_L \kappa_R |D^R(\omega)|^2, \quad (27a)$$

$$R(\omega) = 1 + 2\kappa_L \Im D^R(\omega) + \kappa_L^2 |D^R(\omega)|^2, \quad (27b)$$

$$A(\omega) = -\kappa_L [\kappa |D^R(\omega)|^2 + 2\Im D^R(\omega)]. \quad (27c)$$

Unlike most quantum optics treatments in the literature which deal with $T = 0$ and harmonic degrees of freedom alone, Eqs. 25 and 27 are valid for molecular microcavity Hamiltonians H (Eq. 1) with arbitrary anharmonic molecular degrees of freedom H_{mol} , arbitrary number of molecules N , and the trace in Eq. 25 is carried out with respect to the initial density matrix ρ_0 of the molecular microcavity, $\langle \cdot \rangle = \text{Tr}(\rho_0 \cdot)$ which is assumed to be stationary under evolution with H' .

Eqs. 25 and 27 have been reported in the literature (see [38], Eqs. 15 and 16), although as far as we are aware, their explicit derivation has not been presented anywhere (see also Eq. A4 in [36] although it has a typo of $\frac{1}{2}$). For completeness, Appendix VII provides a self-contained derivation of the latter, listing the assumptions involved.

B. Formulas for $N \rightarrow \infty$

Since the retarded photon Green function (Eq. 25) depends only on the photon *system* (and not on the molecular *bath* degrees of freedom), we can capitalize on the statements of Section II. Replacing $H \rightarrow H^{\text{eff}}$, and working within the RWA, Eq. 10 becomes

$$V^{\text{eff}} = -a \sum_j c_j \sqrt{\frac{\hbar}{2m_j \omega_j}} b_j^\dagger + \text{h.c.} \quad (28)$$

An analytical expression for $D^R(\omega)$ can be obtained for this harmonic surrogate setup; subsection VII C (Appendix) presents explicit derivations. However, to provide additional intuition, here we provide an alternative, possibly simpler derivation. We exploit the profound fact that the response of a harmonic system is *independent* of initial condition of the oscillators (see discussion in VII C 1, Appendix). This means that we might as well take the initial state to be the vacuum $|0\rangle$ of all the effective oscillators and the photon (even if ρ_{mol} does not correspond to $T = 0$!). Then $H^{\text{eff}}|0\rangle = 0$, so Eq. 25 becomes

$$\begin{aligned} D^R(\omega) &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) \langle 0 | a e^{-iH^{\text{eff}}t/\hbar} a^\dagger | 0 \rangle \\ &= \langle 1_{ph} | G^{\text{eff}}(\omega) | 1_{ph} \rangle, \end{aligned} \quad (29)$$

where $|1_{ph}\rangle = a^\dagger |0\rangle$ is the one-photon state, and

$$\begin{aligned} G^{\text{eff}}(\omega) &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) e^{-iH^{\text{eff}}t/\hbar} \\ &= \frac{1}{\omega - H^{\text{eff}}/\hbar} \end{aligned} \quad (30)$$

is the retarded Green function for H^{eff} . Notice that according to Eq. 28, the light-matter coupling V^{eff} only mixes $|1_{ph}\rangle$ with states in the first-excitation manifold of the effective harmonic bath $\{|1_{b_j}\rangle = b_j^\dagger |0\rangle\}$. Thus, within the first excitation manifold, in the basis $\{|1_{ph}\rangle, \{|1_{b_j}\rangle\}\}$, H^{eff} is an arrowhead matrix (that is, $|1_{ph}\rangle$ couples to the first-excitation states $|1_{b_j}\rangle$ but there are no couplings among the latter or between the latter and other states. Defining the zeroth-order retarded Green function $G_0^{\text{eff}} = [\omega - H_0^{\text{eff}}/\hbar]^{-1}$, we can deploy standard Green function identities,

$$\begin{aligned} G_0^{-1} &= G^{-1} + \frac{V}{\hbar} \\ \Rightarrow G_0 G_0^{-1} G &= G_0 G^{-1} G + G_0 \frac{V}{\hbar} G \\ \Rightarrow G &= G_0 + G_0 \frac{V}{\hbar} G \\ &= G_0 + G_0 \frac{V}{\hbar} G_0 + G_0 \frac{V}{\hbar} G_0 \frac{V}{\hbar} G, \end{aligned} \quad (31)$$

to obtain the desired expression [44],

$$\begin{aligned} D^{(R)}(\omega) &= \langle 1_{ph} | G^{\text{eff}}(\omega) | 1_{ph} \rangle \\ &= \frac{1}{(\omega - \omega_{ph} + i\frac{\kappa}{2}) - \Sigma_M}, \end{aligned} \quad (32)$$

TABLE 1: Linear optics of molecular polaritons		
	arbitrary N	$N \rightarrow \infty$
transmission	$T(\omega) = \kappa_L \kappa_R D^R(\omega) ^2$	$T(\omega) = \frac{\kappa_L \kappa_R}{ \omega - \omega_{ph} + i\frac{\kappa}{2} + \chi(\omega) ^2}$
reflection	$A(\omega) = -\kappa_L [\kappa D^R(\omega) ^2 + 2\Im D^R(\omega)]$	$A(\omega) = \frac{2\kappa_L \Im \chi(\omega)}{ \omega - \omega_{ph} + i\frac{\kappa}{2} + \chi(\omega) ^2}$
absorption	$R(\omega) = 1 + 2\kappa_L \Im D^R(\omega) + \kappa_L^2 D^R(\omega) ^2$	$R(\omega) = \frac{ \omega - \omega_{ph} + i\frac{\kappa}{2} + \chi(\omega) ^2 - \kappa_L \kappa_R - 2\kappa_L \Im \chi(\omega)}{ \omega - \omega_{ph} + i\frac{\kappa}{2} + \chi(\omega) ^2}$

Table I.

where Σ_M is the molecular self-energy, which using Eqs. 28 and 20 is found to be minus the molecular susceptibility!

$$\begin{aligned}\Sigma_M &= \lim_{\gamma \rightarrow 0} \sum_j \frac{|\frac{1}{\hbar} V_{ph,1j}^{\text{eff}}|^2}{\omega - \omega_j + i\frac{\gamma}{2}} \\ &= \lim_{\gamma \rightarrow 0^+} \sum_j \frac{|\bar{c}_j|^2}{\omega - \omega_j + i\frac{\gamma}{2}} \\ &= -\chi(\omega).\end{aligned}\quad (33)$$

Note that the molecular susceptibility $\chi(\omega)$ is related to the refractive index of the absorbing material $n(\omega)$ through the relation $n(\omega) = \sqrt{\epsilon_\infty + \chi(\omega)}$, where $n_0 = \sqrt{\epsilon_\infty}$ is the refractive index of the bare cavity, assumed here to be a constant [22]. Eqs. 32 and 33 can be plugged into Eq. 27 to give the compact formulas,

$$T(\omega) = \frac{\kappa_L \kappa_R}{|\omega - \omega_{ph} + i\frac{\kappa}{2} + \chi(\omega)|^2}, \quad (34a)$$

$$A(\omega) = \frac{2\kappa_L \Im \chi(\omega)}{|\omega - \omega_{ph} + i\frac{\kappa}{2} + \chi(\omega)|^2}, \quad (34b)$$

and reflection can be easiest obtained by subtraction, $R(\omega) = 1 - T(\omega) - A(\omega)$. Eq. 34 is the main result

IV. EXAMPLES

In the next subsections we will illustrate the formalism above, and in particular, the use of the formulas in the rightmost column of Table I in concrete examples highlighting effects of temperature, disorder, and optical saturation of the molecular ensemble. Even though the examples are simplified, they should serve as pedagogical tools, highlighting the essence of these effects in the spectra of polaritons. These examples can be readily generalized to account for more realistic molecular details.

A. $N \gg 1$ Two-level systems

We start with the simple example of an ensemble of N two-level atoms coupled to the photon mode via their

of the article. Incidentally, note that we may write the transmission spectrum as a Landauer formula (see for example, Appendix 9B in [41],

$$T(\omega) = \text{Tr}[\Gamma^{(L)} G^{\text{eff}\dagger}(\omega) \Gamma^{(R)} G^{\text{eff}}(\omega)], \quad (35)$$

where $\Gamma^{(i)} = \kappa_i |1_{ph}\rangle \langle 1_{ph}|$ ($i = L, R$). These results state that in the thermodynamic limit ($N \gg 1$), information about the molecular linear susceptibility alone is enough to predict linear response properties of molecular microcavities, thus bypassing simulations where the cavity is explicitly included. Importantly, they have been derived assuming an initially uncorrelated state between light and matter [6], so they might only be useful when SC happens for high-frequency molecular vibrations [45, 46] and electronic excitations [47], but shall be used with caution for collective low-frequency modes, as in Terahertz polaritonics [48, 49], since finite temperature might populate polariton states and create correlations between light and matter. As far as we are aware, these statements were first made in [36, 38], and are consistent with similar claims in [37, 50]. Table I summarizes the main results of this article.

optical transition (Tavis-Cummings model [51]),

$$H_{mol} = \sum_{i=1}^N \hbar \omega_{exc,i} \sigma_i^\dagger \sigma_i, \quad (36a)$$

$$V = -\hbar \lambda a \sum_{i=1}^N \mu_i \sigma_i^\dagger + \text{h.c.}, \quad (36b)$$

where $\sigma^\dagger = |e\rangle \langle g|$ and μ_i is the amplitude of the optical transition. The molecular susceptibility according to Eq. 21 is,

$$\chi(\omega) = - \sum_i \tanh \frac{\beta \hbar \omega_{exc,i}}{2} \left(\frac{|\lambda \mu_i|^2}{\omega - \omega_{exc,i} + i\frac{\gamma}{2}} \right). \quad (37)$$

Let us first assume that all atoms are identical, $\omega_{exc,i} = \omega_{exc}$ and $\mu_i = \mu$. Denoting $g^2 = |\lambda \mu|^2 \tanh \frac{\beta \hbar \omega_{exc}}{2}$, it

follows that $\chi(\omega) = -\frac{N|g|^2}{\omega - \omega_{exc,i} + i\frac{\gamma}{2}}$, which can be inserted into Eq. 34 to give (see Fig. 3 a),

$$T(\omega) = \frac{\kappa_L \kappa_R [(\omega - \omega_{exc})^2 + (\frac{\gamma}{2})^2]}{|(\omega - \omega_{ph} + i\frac{\kappa}{2})(\omega - \omega_{exc} + i\frac{\gamma}{2}) - Ng^2|^2}, \quad (38a)$$

$$A(\omega) = \frac{\kappa_L \gamma Ng^2}{|(\omega - \omega_{ph} + i\frac{\kappa}{2})(\omega - \omega_{exc} + i\frac{\gamma}{2}) - Ng^2|^2}. \quad (38b)$$

Eq. 38 are the standard results for the simplest polariton system with collective light-matter coupling is equal to $\sqrt{N}g$ (see for instance, the Supplementary Information of [28]). Note that the $\beta = 1/k_B \mathcal{T}$ models the relative ground and excited state populations in the two level systems. This is a special case of Eq. 17 for a two-level system, where we can always define a single temperature $\mathcal{T}$.

The renormalization of light-matter coupling g with $\mathcal{T}$ has been previously obtained in [36]. Essentially, owing to optical saturation, Rabi splitting contraction (also known as “phase space filling” in the solid state literature [52]) is expected for high temperatures as compared to $\mathcal{T} \rightarrow 0$, (see Fig. 3b). As $\mathcal{T} \rightarrow \infty$, the light-matter coupling g vanishes because the photon mode coherently adds up the absorption and emission events; neither outside nor inside the cavity do we expect any net absorption. In other words, this transparent molecular medium inside a cavity makes the latter behave as an empty cavity. These effects can be obtained in transient absorption experiments, when optical pumping and dephasing ensues, creating steady-state populations in excited molecular states [53, 54]; for instance, the $\mathcal{T} \rightarrow \infty$ was achieved in [55]. Thus, despite the nonlinear optical nature of transient absorption experiments, we see that the (linear) transmission of the probe upon dephasing and relaxation to the “dark states” obeys the very simple physics outlined in this article, in agreement with the conclusions in [28].

It is also instructive to rederive these results in a more pedestrian way. Solving for the parameters of the surrogate setup,

$$J^{\text{eff}}(\beta, \omega) = \hbar N |g|^2 \delta(\omega_{exc} - \omega), \quad (39a)$$

$$H_{mol}^{\text{eff}} = \hbar \omega_{exc} b^\dagger b, \quad (39b)$$

$$V^{\text{eff}} = -\sqrt{N} |g| (ab^\dagger + \text{h.c.}), \quad (39c)$$

In the $\{|1_{ph}\rangle, |1_b\rangle\}$ photon-exciton basis, we can write the effective Green function,

$$G^{\text{eff}}(\omega) = \begin{bmatrix} \omega - \omega_{ph} + i\frac{\kappa}{2} & \sqrt{N}g \\ \sqrt{N}g & \omega - \omega_{exc} + i\frac{\gamma}{2} \end{bmatrix}^{-1} \\ = \frac{\begin{bmatrix} \omega - \omega_{exc} + i\frac{\gamma}{2} & -\sqrt{N}g \\ -\sqrt{N}g & \omega - \omega_{ph} + i\frac{\kappa}{2} \end{bmatrix}}{(\omega - \omega_{ph} + i\frac{\kappa}{2})(\omega - \omega_{exc} + i\frac{\gamma}{2}) - g^2 N}, \quad (40)$$

as well as the matrices denoting coupling to photonic continua, $\Gamma^{(L)} = \begin{bmatrix} \kappa_L & 0 \\ 0 & 0 \end{bmatrix}$, $\Gamma^{(R)} = \begin{bmatrix} \kappa_R & 0 \\ 0 & 0 \end{bmatrix}$. Applying the Landauer-type formula of Eq. 35, we readily recover Eq. 38a.

As an follow-up to this model, let us now consider the effects of a probability distribution $p(\omega_{exc})$ on the excitation energies ω_{exc} at $T = 0$; then,

$$\chi(\omega) = -N |g|^2 \int d\omega_{exc} \frac{p(\omega_{exc})}{\omega - \omega_{exc} + i\frac{\gamma}{2}}. \quad (41)$$

We consider two types of disorder: Gaussian (Fig. 4 a),

$$p(\omega_{exc}) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(\omega_{exc} - \bar{\omega})^2 / 2\sigma^2} \quad (42)$$

and Lorentzian (Fig. 4 b),

$$p(\omega_{exc}) = \frac{1}{\pi} \frac{(\sigma/2)^2}{(\omega_{exc} - \bar{\omega})^2 + (\sigma/2)^2}. \quad (43)$$

In Fig. 4, computed using Eq. 42, we demonstrate the line narrowing in the polaritonic absorption with increasing collective light-matter coupling for an ensemble with Gaussian disorder. This phenomenon relies on the absorption lineshape being primarily governed by the homogeneous linewidth of the molecules and the cavity for $g\sqrt{N} \gg \sigma$. This is consistent with the seminal work of [56], which has been recently revisited in many other studies [36, 57–65].

B. $N \gg 1$ Two-level molecules

We generalize the Tavis-Cummings above and add one vibration coupled to each electronic transition,

$$H_{mol} = \sum_{i=1}^N \hbar \omega_{exc,i} \sigma_i^\dagger \sigma_i + \hbar \omega_v [b_i^\dagger b_i - \sqrt{S} \sigma_i^\dagger \sigma_i (b_i^\dagger + b_i)] \\ V = -\hbar \lambda (a + a^\dagger) \sum_{i=1}^N \mu_i (\sigma_i^\dagger + \sigma_i) + \text{h.c.}$$

Here, $b_i^\dagger$ (b_i) is the creation (annihilation) operator for a vibrational excitation in a high-frequency ω_v harmonic

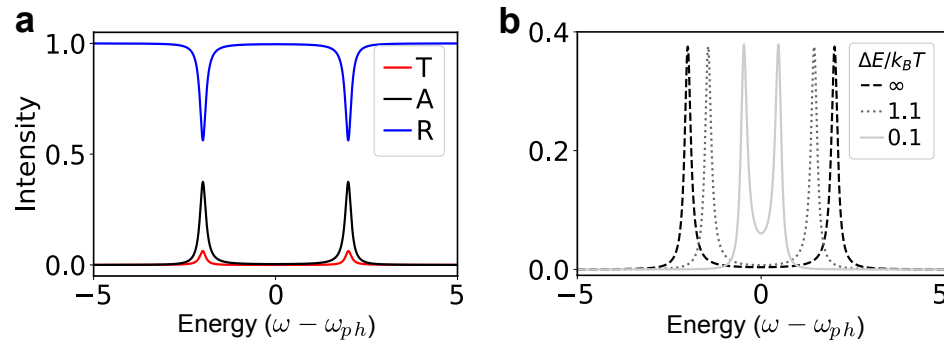


Figure 3. (a) Transmission, reflection, and absorption for an ensemble of identical two-level systems, calculated with Eq. 38. (b) Absorption for an ensemble of identical two-level systems at different temperatures demonstrating Rabi splitting contraction. The spectra have been computed for resonant light and matter energies $\omega_{ph} = \omega_{exc}$, cavity and molecule decay rates $\kappa = 0.1$, $\gamma = 0.3$, and collective light-matter coupling $\sqrt{N}g = 2$ (arbitrary frequency units throughout all figures).

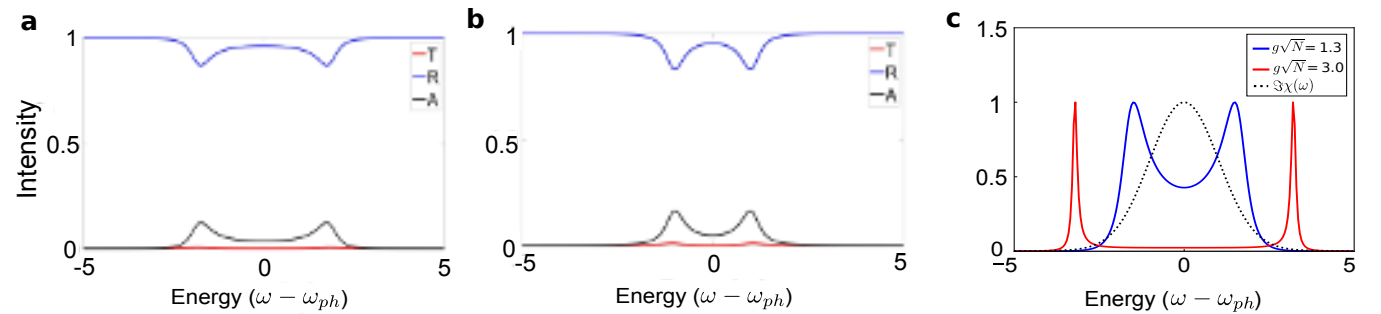


Figure 4. Transmission, reflection, and absorption for an ensemble of (a) Gaussian energy-distributed two-level systems ($\sigma = 1$, see Eq. 42) and (b) Lorentzian energy-distributed two-level systems ($\sigma = 1$, see Eq. 43) obtained using Eqs. 34. We show the examples with resonant light and matter energies $\omega_{ph} = \omega_{exc}$, cavity and molecule decay rates $\kappa = \gamma = 0.1$, and collective light-matter coupling $\sqrt{N}g = 1.5$. (c) Normalized polariton absorption for Gaussian disorder $\sigma = 1$ using Eq. 42, illustrating the variation of the linewidth with collective light-matter coupling strength, $g\sqrt{N} = 1.3$ and 3.0 . The bare Gaussian molecular absorption is also shown for completeness. For $g\sqrt{N} \gg \sigma$, we see linewidth narrowing dominated by homogeneous broadening κ, γ , consistent with the observations of Houdré [56].

mode of the i -th molecule. The electron-vibration coupling is characterized by the so-called Huang-Rhys parameter S . This system has been previously studied by [38] (see their Fig. 8). To get started, we consider a $T = 0$ ensemble of N identical molecules, $\omega_{exc,i} = \omega_{exc}$, $\mu_i = \mu$. Then, defining $g^2 = |\lambda\mu|^2$ the susceptibility is,

$$\chi(\omega) = - \sum_m \frac{Ng^2 |\langle m'|0 \rangle|^2}{\omega - (\omega_{exc} - S\omega_v + m\omega_v) + i\frac{\gamma}{2}}. \quad (44)$$

Here, $|\langle m'|0 \rangle|^2 = e^{-S} \frac{S^m}{m!}$ is the Franck-Condon factor for the $|0\rangle \rightarrow |m'\rangle$ vibronic transition, and the vertical transition is at frequency ω_{exc} , which contains $S\omega_v$ vibrational quanta. The spectra coincides with that obtained using the Collective dynamics Using Truncated Equations (CUT-E) method developed to address the quantum dynamics of ensembles of N complex molecules coupled to a cavity, in the limit when $O(N^{-k})$ effects ($k \geq 1$) are ignored [50].

C. $N \gg 1$ Three-level systems

We conclude our examples by considering an illustrative ensemble of N three-level systems placed in a cavity. The three optical transitions couple to the cavity-photon mode. The Hamiltonian is given as

$$H_{mol} = \sum_{y=1}^3 \sum_{i=1}^N \hbar\omega_{y,i} |y_i\rangle \langle y_i|, \quad (45a)$$

$$V = -\hbar\lambda a \sum_{y,z=1,y \neq z}^3 \sum_{i=1}^N \mu_{zy,i} |z_i\rangle \langle y_i|, \quad (45b)$$

where $\mu_{zy,i} = \langle z_i | \mu | y_i \rangle$ is the amplitude for the $|y\rangle \rightarrow |z\rangle$ transition in the i^{th} molecule. When all the molecules in the ensemble are identical, $\omega_{zy,i} = \omega_{zy}$ and $\mu_{zy,i} = \mu_{zy}$ for all i , the susceptibility $\chi(\omega)$ in Eq. 21b reads

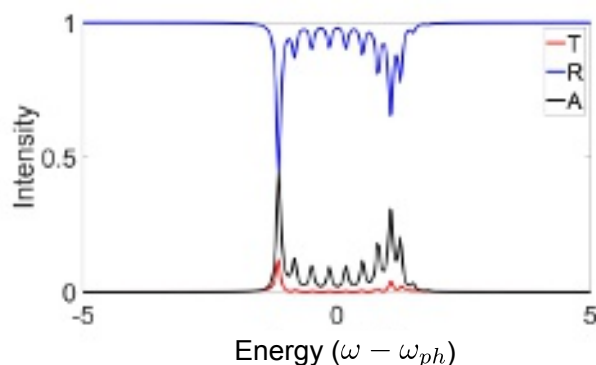


Figure 5. Transmission, reflection, and absorption for an ensemble of identical two-level systems with vibronic coupling obtained using Eq. 34. We show an example with resonant light and matter energies $\omega_{ph} = \omega_{exc}$, cavity and molecule decay rates $\kappa = \gamma = 0.1$, collective light-matter coupling $\sqrt{N}g = 1$ (arbitrary frequency units), $\omega_v = 0.3$, and $S = 3$.

$$\chi(\omega) = - \sum_{y,z=1}^3 (p_y - p_z) \frac{N\lambda^2 |\mu_{zy}|^2}{\omega - \omega_{zy} + i\frac{\gamma}{2}}. \quad (46)$$

The transmission, absorption and reflection spectra of the setup have been shown in Fig.6 for different population ratios. For $p_1 > p_2 > p_3$, we see the four polariton peaks owing to the three optical transitions at distinct frequencies coupling to the cavity (Fig.6 a). Optical saturation of one of the transitions ($p_y = p_z$) causes those transitions to become transparent to the cavity (the inverse temperature for this transition becomes $\beta_{yz}^{\text{eff}} = 0$) (see Eq. 17 and the discussion about Rabi splitting contraction in Subsection IV A), and show three polariton peaks in the spectra (Fig. 6b). For $p_1 = p_2 = p_3$ all transitions are saturated, and the system behaves like an empty cavity (Fig. 6c). If p_i do not obey a Boltzmann distribution, these states can in principle be obtained in optical pumping experiments after dephasing decouples light and matter; if they do (as in the last case), they can be also be obtained by tuning the surroundings at the corresponding temperature $\mathcal{T}$.

V. CONCLUSION

In this article, we have treated the polariton problem as a quantum impurity model where the photon is the impurity coupled to the optical transitions of N molecules. In the large N limit, we have shown that the photon Green's function can be trivially obtained with the linear susceptibility $\chi(\omega)$ of the bare molecule (which in turn, owing to Kramers Kronig relations, can be obtained solely from bare molecular absorption spectra), bypassing a costly simulation of N explicit molecules coupled to a cavity (Eq. 34). This result is quite general and

is consistent with the success with which transfer matrix methods in classical optics are used to model polariton spectra. Importantly, it holds for arbitrary initial states so long as the light and the matter are decoupled and stationary, providing us with a very simple tool to understand a wide scope of phenomena (Section IV), ranging from Rabi splitting features in idealized ensembles, but also complex lineshapes involving optical saturation, disorder, vibronic coupling, and nonequilibrium stationary states such as those that emerge upon optical pumping (*i.e.*, the many situations where polaritons have relaxed to incoherent “dark states,” yet they have not fully thermalized, as in ultrafast experiments).

For small N , $\chi(\omega)$ as the molecular information is not enough to compute a polariton spectrum. However, the latter can still be expressed in terms of a photon Green's function, as per Kubo linear response formalism (Eq. 27). These expressions have been featured in previous studies [36, 38], but their derivation was not explicitly provided. We do so in the Appendix and discuss their simplifications in the harmonic ($N \rightarrow \infty$) regime.

Finally, the fact that polaritons, regarded as a quantum impurity problems, are simple when $N \rightarrow \infty$, is a possibly counterintuitive yet intriguing observation [66–71]. In fact, recent work explores this fact to develop a mean-field approach to polariton dynamics [72]. In another work, we have recently shown that there is a hierarchy of timescales that allows for the efficient simulation of molecular polaritons when $N \gg 1$ but not infinite (Collective dynamics Using Truncated Equations, CUT-E [50, 73]), giving rise to $O(N^{-k})$ rates that account for the finite size of the molecular ensemble, very much in the spirit of $1/N$ expansions in other fields including quantum field theory and quantum chemistry. Understanding the class of quantum dynamics problems that afford similar strategies is a fascinating direction to be explored in the near future.

VI. ACKNOWLEDGEMENTS

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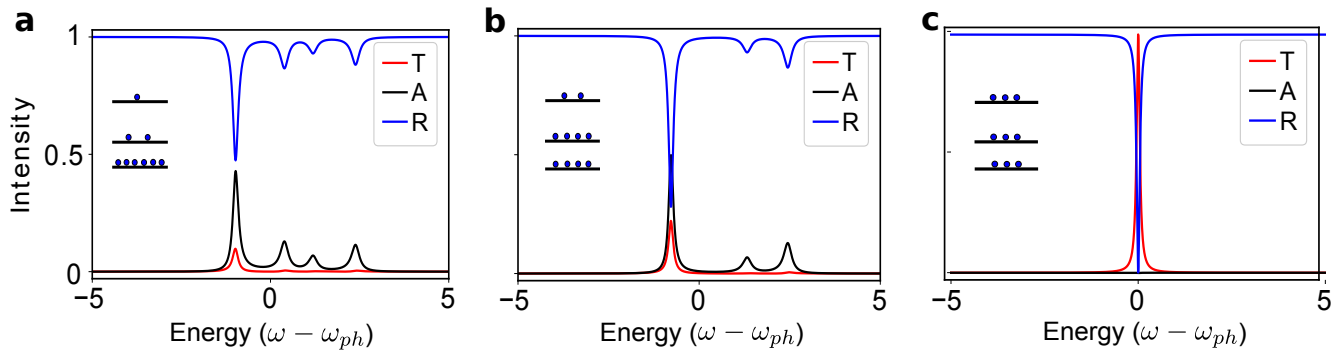


Figure 6. Transmission, reflection, and absorption for an ensemble of identical three-level systems for (a) $p_1 = 0.7$, $p_2 = 0.2$, $p_3 = 0.1$, (b) $p_1 = 0.48$, $p_2 = 0.48$, $p_3 = 0.04$, (c) $p_1 = p_2 = p_3$, obtained using Eq. 34. The spectra have been computed for the parameters, $\omega_{ph} = \omega_{12}$, $\omega_{23} = 2\omega_{12}$, cavity and molecule decay rates $\kappa = 0.1$, $\gamma = 0.3$, collective light-matter coupling $\sqrt{N}g = 1$ (arbitrary frequency units).

VII. APPENDIX: DERIVATION OF SPECTROSCOPIC OBSERVABLES

We will assume the RWA throughout this Appendix.

A. Input-output theory

To keep this manuscript self-contained, we derive formulas for linear spectroscopy of cavity polaritons using input-output (IO) theory [23–27]. We couple the molecular microcavity (Eq. 1) with left and right radiative continua via the photon mode,

$$H_{\text{total}} = H + H_L + H_R, \quad (\text{A1})$$

where

$$H_K = \int_0^\infty d\omega' \hbar \omega' b^{K\dagger}(\omega') b^K(\omega') + \left[\hbar \frac{\sqrt{\kappa_K}}{\sqrt{2\pi}} \int_0^\infty d\omega' b^{K\dagger}(\omega') a + \text{h.c.} \right] \quad (\text{A2})$$

for $K = L, R$ only feature RWA terms and

$$[b^K(\omega), b^{K'}(\omega')] = [b^{K\dagger}(\omega), b^{K'\dagger}(\omega')] = 0, \quad (\text{A3a})$$

$$[b^K(\omega), b^{K'\dagger}(\omega')] = \delta_{KK'} \delta(\omega - \omega'). \quad (\text{A3b})$$

For convenience, we will now derive some results in the Heisenberg picture (corresponding to evolution with respect to H_{total}), with the corresponding operators labeled by the subscript H, *e.g.*, $a_H(t) = e^{-iH_{\text{total}}(t-t_{\text{in}})/\hbar} a e^{iH_{\text{total}}(t-t_{\text{in}})/\hbar}$. Schrödinger picture operators will continue to be indicated without an explicit subscript. The Equation of Motion (EoM) for the cavity photon is

$$\frac{\partial a_H(t)}{\partial t} = -\frac{i}{\hbar} [a_H(t), H_H(t)] - i \sum_{K=L,R} \sqrt{\frac{\kappa_K}{2\pi}} \int_0^\infty d\omega' b_H^K(\omega')(t), \quad (\text{A4})$$

Similarly, the corresponding EoM for the bath mode $b_H^K(\omega)(t)$ is

$$\begin{aligned} \frac{\partial b_H^K(\omega)(t)}{\partial t} &= -i\omega b_H^K(\omega) - i\sqrt{\frac{\kappa_K}{2\pi}} a_H(t) \\ \Rightarrow \frac{\partial [b_H^K(\omega)(t) e^{i\omega t}]}{\partial t} &= -i\sqrt{\frac{\kappa_K}{2\pi}} a_H(t) e^{i\omega t}. \end{aligned} \quad (\text{A5})$$

Defining $t_{\text{in}} < t$ and $t_{\text{out}} > t$, we can integrate Eq. A5 to obtain,

$$b_H^K(\omega)(t) = b_H^K(\omega)(t_{\text{in}}) e^{-i\omega(t-t_{\text{in}})} - i\sqrt{\frac{\kappa_K}{2\pi}} \int_{t_{\text{in}}}^t dt' a_H e^{-i\omega(t-t')}, \quad (\text{A6a})$$

$$b_H^K(\omega)(t) = b_H^K(\omega)(t_{\text{out}}) e^{-i\omega(t-t_{\text{out}})} + i\sqrt{\frac{\kappa_K}{2\pi}} \int_t^{t_{\text{out}}} dt' a_H e^{-i\omega(t-t')}. \quad (\text{A6b})$$

Let us feed Eq. A6 for $K = L$ into Eq. A4. By approximating $\int_0^\infty d\omega' \approx \int_{-\infty}^\infty d\omega'$,

$$\begin{aligned} \frac{\partial a_H(t)}{\partial t} &= -\frac{i}{\hbar} [a_H(t), H_H(t)] - i\sqrt{\frac{\kappa_R}{2\pi}} \int_0^\infty d\omega' b_H^R(\omega')(t) \\ &\quad - \frac{\kappa_L}{2} a_H(t) - \sqrt{\kappa_L} b_{\text{in},H}^L(t) \end{aligned} \quad (\text{A7a})$$

$$\begin{aligned} &= -\frac{i}{\hbar} [a_H(t), H_H(t)] - i\sqrt{\frac{\kappa_R}{2\pi}} \int_0^\infty d\omega' b_H^R(\omega')(t) \\ &\quad + \frac{\kappa_L}{2} a_H(t) - \sqrt{\kappa_L} b_{\text{out},H}^L(t). \end{aligned} \quad (\text{A7b})$$

where

$$b_{\text{in,H}}^K(t) = \frac{i}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega' b_{\text{H}}^K(\omega)(t_{\text{in}}) e^{-i\omega(t-t_{\text{in}})}, \quad (\text{A8a})$$

$$b_{\text{out,H}}^K(t) = \frac{i}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega' b_{\text{H}}^K(\omega)(t_{\text{out}}) e^{-i\omega(t-t_{\text{out}})}. \quad (\text{A8b})$$

By comparing Eqs. A7a and A7b, we obtain the IO relations for the left continuum,

$$b_{\text{out,H}}^L(t) - b_{\text{in,H}}^L(t) = \sqrt{\kappa_L} a_{\text{H}}(t). \quad (\text{A9a})$$

It is clear that repeating the procedure of Eqs. A7–A8 with $K = R$ yields the analogous IO relations for the right continuum,

$$b_{\text{out,H}}^R(t) - b_{\text{in,H}}^R(t) = \sqrt{\kappa_R} a_{\text{H}}(t). \quad (\text{A9b})$$

For completeness, we can also express Eq. A4 in terms of the IO operators of both continua,

$$\frac{\partial a_{\text{H}}(t)}{\partial t} = -\frac{i}{\hbar} [a_{\text{H}}(t), H_{\text{H}}(t)] + \sum_K \left[-\frac{\kappa_K}{2} a_{\text{H}}(t) - \sqrt{\kappa_K} b_{\text{in,H}}^K(t) \right] \quad (\text{A10a})$$

$$= -\frac{i}{\hbar} [a_{\text{H}}(t), H_{\text{H}}(t)] + \sum_K \left[\frac{\kappa_K}{2} a_{\text{H}}(t) - \sqrt{\kappa_K} b_{\text{out,H}}^K(t) \right]. \quad (\text{A10b})$$

Assuming that the density matrix at $t = t_{\text{in}}$ is a product state between the molecular microcavity and the continua,

$$\rho_{\text{total}}(t_{\text{in}}) = \rho_L(t_{\text{in}}) \otimes \rho(t_{\text{in}}) \otimes \rho_R(t_{\text{in}}), \quad (\text{A11})$$

and that the driving occurs only from the left continuum,

$$\langle b_{\text{in,H}}^L(t) \rangle \neq 0, \quad (\text{A12a})$$

$$\langle b_{\text{in,H}}^R(t) \rangle = 0, \quad (\text{A12b})$$

we get, after tracing over the continua,

$$\frac{\partial a_{\text{H}}(t)}{\partial t} = -\frac{i}{\hbar} [a_{\text{H}}(t), H_{\text{H}}(t)] - \frac{\kappa}{2} a_{\text{H}}(t) - \sqrt{\kappa_L} \langle b_{\text{in,H}}^L(t) \rangle, \quad (\text{A13})$$

where $\kappa = \kappa_L + \kappa_R$. Eq. A13 can be rewritten as,

$$\frac{\partial a_{\text{H}}(t)}{\partial t} = -\frac{i}{\hbar} [a_{\text{H}}(t), \tilde{H}_{\text{H}}(t)], \quad (\text{A14})$$

which allows us to conclude that the effective time-dependent Hamiltonian (in the Schrödinger picture) governing the molecular microcavity is

$$\tilde{H}(t) = H' + H_{\text{int}}(t). \quad (\text{A15a})$$

In the absence of drive, the molecular microcavity obeys the effective non-Hermitian Hamiltonian

$$H' = H - i\frac{\hbar\kappa}{2} a^\dagger a, \quad (\text{A15b})$$

while the time-dependent drive of the cavity due to light coupling from the left hand side is

$$H_{\text{int}}(t) = -i\hbar\sqrt{\kappa_L} \langle b_{\text{in,H}}^L(t) \rangle a^\dagger + \text{h.c.} \quad (\text{A15c})$$

Equipped with this formalism, we are interested in computing the following spectroscopic observables,

$$T(\omega) = \frac{|\langle b_{\text{out,H}}^R(\omega) \rangle|^2}{|\langle b_{\text{in,H}}^L(\omega) \rangle|^2} = \frac{\kappa_R |\langle a_{\text{H}}(\omega) \rangle|^2}{|\langle b_{\text{in,H}}^L(\omega) \rangle|^2}, \quad (\text{A16a})$$

$$R(\omega) = \frac{|\langle b_{\text{out,H}}^L(\omega) \rangle|^2}{|\langle b_{\text{in,H}}^L(\omega) \rangle|^2} = \frac{|\langle \sqrt{\kappa_L} a_{\text{H}}(\omega) + b_{\text{in,H}}^L(\omega) \rangle|^2}{|\langle b_{\text{in,H}}^L(\omega) \rangle|^2}, \quad (\text{A16b})$$

$$A(\omega) = 1 - T(\omega) - R(\omega), \quad (\text{A16c})$$

where the traces above are carried out with respect to the initial state (Eq. A11), $\langle \cdot \rangle = \text{Tr}[\cdot \rho_{\text{total}}(t_{\text{in}})]$, and in particular, $\langle a_{\text{H}}(\omega) \rangle = \text{Tr}[a_{\text{H}}(\omega) \rho(t_{\text{in}})]$ only depends on the initial state of the molecular microcavity. We have also used Eq. A12 and the Fourier transform convention in Eq. 23. Eq. A16 reveals that all the relevant spectroscopic observables can be obtained once $\langle a_{\text{H}}(\omega) \rangle$ is known. We now show two scenarios where $\langle a_{\text{H}}(\omega) \rangle$ can be easily computed.

B. $\langle a_{\text{H}}(\omega) \rangle$ from Kubo linear response

1. Derivation of transmission, reflection, and absorption formulas

Hereafter we set $t_{\text{in}} = 0$. In the general case, H contains anharmonic terms, so the evaluation of $\langle a_{\text{H}}(\omega) \rangle$ cannot be performed exactly. Instead, we can carry out a perturbation expansion in H_{int} for each of the Heisenberg operators in Eq. A14 and solve for $a_{\text{H}}^{(n)}(t)$ up to lowest nonvanishing order n .

Starting at zeroth-order $O(H_{\text{int}}^0)$,

$$\frac{\partial a_{\text{H}}^{(0)}(t)}{\partial t} - \frac{i}{\hbar}[H', a_{\text{H}}^{(0)}(t)] = 0 \quad (\text{B1})$$

can be solved by,

$$a_{\text{H}}^{(0)}(t) = e^{iH't/\hbar} a e^{-iH't/\hbar}. \quad (\text{B2})$$

Recall our assumption that the initial molecular micro-cavity state is a product state between photon and molecular degrees of freedom (see 6), $\rho(0) = \rho_{\text{ph}} \otimes \rho_{\text{mol}}$,

$$e^{-iH't/\hbar} \rho(0) \xrightarrow{t \rightarrow \infty} |0\rangle \langle \varphi_{\text{ph}}| \otimes \rho_{\text{mol}}. \quad (\text{B3})$$

where $|\varphi_{\text{ph}}\rangle$ is a photonic state. Then,

$$\begin{aligned} \langle a_{\text{H}}^{(0)}(t) \rangle &= \text{Tr}[e^{iH't/\hbar} a e^{-iH't/\hbar} \rho(t_{\text{in}})] \\ &\xrightarrow{t \rightarrow \infty} 0, \end{aligned} \quad (\text{B4})$$

which makes sense since any transient photonic amplitude will vanish due to photon escape.

Similarly, at $O(H_{\text{int}})$, we have

$$\frac{\partial a_{\text{H}}^{(1)}(t)}{\partial t} - \frac{i}{\hbar}[H', a_{\text{H}}^{(1)}(t)] = -\sqrt{\kappa_L} \langle b_{\text{in},\text{H}}^L(t) \rangle. \quad (\text{B5})$$

This is a first order inhomogeneous differential equation that can be solved with Green's function methods. Defining

$$G^R(t) = \Theta(t)[a_{\text{H}}^{(0)}(t), a_{\text{H}}^{(0)\dagger}(0)], \quad (\text{B6})$$

which solves,

$$\frac{\partial G^R(t-t')}{\partial t} - \frac{i}{\hbar}[H', G^R(t-t')] = \delta(t-t'). \quad (\text{B7})$$

where we need an additional assumption: the trace is performed over an initial state $\rho(t_{\text{in}})$ that is stationary with respect to H' ; thus, it contains no photons. We readily obtain the Kubo linear response formula,

$$\begin{aligned} \langle a^{(1)}(t) \rangle &= -\sqrt{\kappa_L} \int_{-\infty}^{\infty} dt_1 \langle b_{\text{in}}^L(t-t_1) \rangle D^R(t_1) \\ &= -\sqrt{\kappa_L} \int_{-\infty}^{\infty} dt_1 \langle b_{\text{in}}^L(t-t_1) \rangle \Theta(t_1) \langle [a_{\text{H}}^{(0)}(t_1), a^\dagger] \rangle. \end{aligned} \quad (\text{B8})$$

where $D^R(t) = \langle G^R(t) \rangle$ is the retarded Green's function. Importantly, Eq. B8 has the form of a convolution,

$$\langle a(\omega) \rangle = \langle a_{\text{H}}^{(1)}(\omega) \rangle = -i\sqrt{\kappa_L} \langle b_{\text{in},\text{H}}^L(\omega) \rangle D^R(\omega), \quad (\text{B9})$$

where $D^R(\omega)$, according to our Fourier transform (23) convention, reads

$$D^R(\omega) = -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) \langle [e^{iH't/\hbar} a e^{-iH't/\hbar}, a^\dagger] \rangle. \quad (\text{B10})$$

Incidentally, given the non-Hermitian nature of H' due to photon leakage, $\rho(0)$ cannot contain photons, so one of the terms in the commutator is superfluous and the final propagator can be replaced, $H' \rightarrow H$,

$$D^R(\omega) = -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) \langle e^{iHt/\hbar} a e^{-iHt/\hbar} a^\dagger \rangle. \quad (\text{B11})$$

Eq. B11 can be fed into Eq. B9. Using Eq. A9, A12, and A16 gives [36, 38],

$$T(\omega) = \kappa_L \kappa_R |D^R(\omega)|^2, \quad (\text{B12a})$$

$$R(\omega) = 1 + 2\kappa_L \Im D^R(\omega) + \kappa_L^2 |D^R(\omega)|^2, \quad (\text{B12b})$$

$$A(\omega) = -\kappa_L [\kappa |D^R(\omega)|^2 + 2\Im D^R(\omega)]. \quad (\text{B12c})$$

2. A "Landauer" formula

We now re-express the photon retarded Green function (Eq. B10) as,

$$\begin{aligned} D^R(\omega) &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) \langle [a(t), a^\dagger] \rangle \\ &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) \langle \mathcal{A}(t) a^\dagger \rangle \\ &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) \langle e^{i\mathcal{L}'t/\hbar} \mathcal{A} e^{-i\mathcal{L}'t/\hbar} a^\dagger \rangle \\ &= \langle \mathcal{AG}(\omega) a^\dagger \rangle, \end{aligned} \quad (\text{B13})$$

where we have introduced Liouville space operators $\mathcal{A}(t) = [a(t), \cdot]$ and $\mathcal{L}' = [H', \cdot]$ (see [33] Chapter 3). We also assumed stationarity of $\rho(0)$ under evolution with respect to $\mathcal{L}'$. Furthermore, we introduced $\mathcal{G}(\omega)$ as the frequency domain retarded Green function

$$\mathcal{G}(\omega) = \frac{1}{\omega - \mathcal{L}'/\hbar}. \quad (\text{B14})$$

Plugging Eq. B13 into Eq. A16a, we obtain a Liouville-space "Landauer" formula for transmission (see for instance, Appendix 9B in [41]),

$$\begin{aligned} T(\omega) &= \kappa_L \kappa_R |D^R(\omega)|^2 \\ &= \kappa_L \kappa_R |\langle \mathcal{AG}(\omega) a^\dagger \rangle|^2. \end{aligned} \quad (\text{B15})$$

C. $\langle a_H(\omega) \rangle$ from harmonic degrees of freedom

As argued in Section II, when the number of molecules $N \rightarrow \infty$, the anharmonic molecular degrees of freedom can be replaced by an effective harmonic bath. Then, the effective molecular microcavity Hamiltonian with photon loss and RWA reads (Eqs. 7–10),

$$H^{\text{eff}} = \hbar(\omega_{ph} - i\frac{\kappa}{2})a^\dagger a + \sum_j \hbar\omega_j b_j^\dagger b_j - \left[a \sum_j \hbar\bar{c}_j b_j^\dagger + \text{h.c.} \right]. \quad (\text{C1})$$

The conclusions from this part VII C of the Appendix will clearly also hold for idealized harmonic Hamiltonians H' , where the $N \rightarrow \infty$ restriction is not needed.

1. $\langle a_H(\omega) \rangle$ from EoM

$\langle a_H(\omega) \rangle$ can be solved exactly for Eq. C1. Using it to evaluate the $[a, H^{\text{eff}}]$ commutator in Eq. A13 yields,

$$\frac{\partial a_H(t)}{\partial t} = -i(\omega_{ph} - i\frac{\kappa}{2})a_H(t) + i\bar{c}_j^* b_{j,H} - \sqrt{\kappa_L} b_{\text{in},H}^L(t) \quad (\text{C2a})$$

$$\Rightarrow -i\omega' a_H(\omega) = -i(\omega_{ph} - i\frac{\kappa}{2})a_H(\omega) + \sum_j i\bar{c}_j^* b_{j,H}(\omega) - \sqrt{\kappa_L} b_{\text{in},H}^L(\omega). \quad (\text{C2b})$$

The corresponding EoM for each of the $b_{j,H}(t)$ is given by,

$$\frac{\partial b_{j,H}(t)}{\partial t} = -i(\omega_j - i\frac{\gamma}{2})b_{j,H}(t) + i\bar{c}_j a_H(t) \quad (\text{C3a})$$

$$\Rightarrow -i\omega b_{j,H}(\omega) = -i(\omega_j - i\frac{\gamma}{2})b_{j,H}(\omega) + i\bar{c}_j a_H(\omega). \quad (\text{C3b})$$

Solving Eq. C3b for $b_{j,H}(\omega)$ and plugging the result into Eq. C2b yields the result,

$$\langle a_H(\omega) \rangle = \frac{-i\sqrt{\kappa_L} \langle b_{\text{in},H}^L(\omega) \rangle}{(\omega - \omega_{ph} + i\frac{\kappa}{2} - \Sigma_M)} \quad (\text{C4})$$

where the molecular self-energy is shown to be minus the linear susceptibility of the molecules,

$$\Sigma_M = \sum_j \frac{|\bar{c}_j|^2}{\omega' - \omega_j + i\frac{\gamma}{2}} = -\chi(\omega). \quad (\text{C5})$$

As expected for a harmonic system, its response to driving is linear, in this case, proportional to $\sqrt{\frac{\kappa}{2}} \langle b_{\text{in},H}^L(\omega) \rangle$. Interestingly, $\langle a_H(\omega) \rangle$ is independent of the initial quantum state of the effective harmonic oscillators. However, recall that the information about the initial thermal state of the real anharmonic degrees of freedom is hidden in $\{\bar{c}_j\}$.

2. $\langle a_H(\omega) \rangle$ from Kubo formula

We now show that the Kubo formula in Eq. B9, despite being derived under a perturbation theory and initial stationary state $\rho(t_{\text{in}})$, is not an approximation, but rather gives the exact response if the molecular degrees of freedom can be treated as harmonic. We invoke the analogous identity of Eq. 31 but in Liouville space (to unclutter notation, we drop the ω argument hereafter when it is clear) to expand Eq. B13,

$$\mathcal{G} = \mathcal{G}_0 + \mathcal{G}_0 \frac{\mathcal{V}}{\hbar} \mathcal{G}_0 + \mathcal{G}_0 \frac{\mathcal{V}}{\hbar} \mathcal{G}_0 \frac{\mathcal{V}}{\hbar} \mathcal{G}, \quad (\text{C6})$$

where the non-interacting Green function $\mathcal{G}_0$ corresponds to $\mathcal{L}_0 = [H'_0, \cdot] = [H_0 - i\hbar\frac{\kappa}{2}a^\dagger a, \cdot]$. This identity implies,

$$\mathcal{A}\mathcal{G}a^\dagger = \mathcal{A}\mathcal{G}_0a^\dagger + \mathcal{A}\mathcal{G}_0\frac{\mathcal{V}}{\hbar}\mathcal{G}_0a^\dagger + \mathcal{A}\mathcal{G}_0\frac{\mathcal{V}}{\hbar}\mathcal{G}_0\frac{\mathcal{V}}{\hbar}\mathcal{G}a^\dagger. \quad (\text{C7})$$

Terms involving $\mathcal{G}_0$ can be readily and explicitly evaluated in Hilbert space,

$$\begin{aligned} \mathcal{A}\mathcal{G}_0^{\text{eff}}a^\dagger &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) [e^{iH'_0 t/\hbar} a e^{-iH'_0 t/\hbar}, a^\dagger] \\ &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) [a, a^\dagger] e^{-i(\omega'_{ph} - i\kappa/2)t} \\ &= \frac{1}{\omega - \omega_{ph} + i\frac{\kappa}{2}}. \end{aligned} \quad (\text{C8})$$

$$\begin{aligned} \mathcal{B}_j \mathcal{G}_0^{\text{eff}} b_{j'}^\dagger &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) [e^{iH'_0 t/\hbar} b_j e^{-iH'_0 t/\hbar}, b_{j'}^\dagger] \\ &= -i \int_{-\infty}^{\infty} dt e^{i\omega t} \Theta(t) [b_j, b_{j'}^\dagger] e^{-i(\omega'_{ph} - i\gamma/2)t} \\ &= \frac{\delta_{jj'}}{\omega - \omega_j + i\frac{\gamma}{2}}. \end{aligned} \quad (\text{C9})$$

The second term in the sum of Eq. C7 can be shown to vanish identically given that V^{eff} couples photon with matter degrees of freedom (but not matter with matter or photon with photon),

$$\begin{aligned}
 & \mathcal{A}\mathcal{G}_0^{\text{eff}} \frac{\mathcal{V}^{\text{eff}}}{\hbar} \mathcal{G}_0^{\text{eff}} a^\dagger \\
 &= \mathcal{A}\mathcal{G}_0^{\text{eff}} (a^\dagger \mathcal{A}) \frac{\mathcal{V}^{\text{eff}}}{\hbar} (a^\dagger \mathcal{A}) \mathcal{G}_0^{\text{eff}} a^\dagger \\
 &= (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) (\mathcal{A} \frac{\mathcal{V}^{\text{eff}}}{\hbar} a^\dagger) (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) \\
 &= \left(\frac{1}{\omega - \omega_{ph} + i\frac{\kappa}{2}} \right)^2 \mathcal{A} \frac{\mathcal{V}^{\text{eff}}}{\hbar} a^\dagger \\
 &= \left(\frac{1}{\omega - \omega_{ph} + i\frac{\kappa}{2}} \right)^2 \mathcal{A} \left[\frac{\mathcal{V}^{\text{eff}}}{\hbar}, a^\dagger \right] \\
 &= \left(\frac{1}{\omega - \omega_{ph} + i\frac{\kappa}{2}} \right)^2 \left[a, -\sum_j \bar{c}_j b_j^\dagger \right] \\
 &= 0.
 \end{aligned} \tag{C10}$$

where we inserted two “minus” resolutions of the identity in the second line,

$$\begin{aligned}
 \mathcal{A}a^\dagger &= [a, a^\dagger] \\
 &= 1
 \end{aligned} \tag{C11a}$$

$$\begin{aligned}
 \Rightarrow a^\dagger \mathcal{A} &= (\mathcal{A}^\dagger a)^\dagger \\
 &= [a^\dagger, a]^\dagger \\
 &= -1
 \end{aligned} \tag{C11b}$$

Finally, the third term in the sum of Eq. C7 is non-zero and leads to the self-energy of the photon due to its coupling to the matter degrees of freedom,

$$\begin{aligned}
 & \mathcal{A}\mathcal{G}_0^{\text{eff}} \frac{\mathcal{V}^{\text{eff}}}{\hbar} \mathcal{G}_0^{\text{eff}} \frac{\mathcal{V}^{\text{eff}}}{\hbar} \mathcal{G}_0^{\text{eff}} a^\dagger \\
 &= \mathcal{A}\mathcal{G}_0^{\text{eff}} (a^\dagger \mathcal{A}) \frac{\mathcal{V}^{\text{eff}}}{\hbar} \mathcal{G}_0^{\text{eff}} \frac{\mathcal{V}^{\text{eff}}}{\hbar} (a^\dagger \mathcal{A}) \mathcal{G}_0^{\text{eff}} a^\dagger \\
 &= (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) \mathcal{A} \frac{\mathcal{V}^{\text{eff}}}{\hbar} \mathcal{G}_0^{\text{eff}} \left(\frac{\mathcal{V}^{\text{eff}}}{\hbar} a^\dagger \right) (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) \\
 &= -\sum_j (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) (\mathcal{A} \frac{\mathcal{V}^{\text{eff}}}{\hbar} b_j^\dagger) \mathcal{B}_j \mathcal{G}_0^{\text{eff}} (-\bar{c}_j b_j^\dagger) (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) \\
 &= -\sum_j (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) \mathcal{A} (-\bar{c}_j a^\dagger) \mathcal{B}_j \mathcal{G}_0^{\text{eff}} (-\bar{c}_j b_j^\dagger) (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) \\
 &= -\left(\frac{1}{\omega - \omega_{ph} + i\frac{\kappa}{2}} \right) \sum_j \left(\frac{|\bar{c}_j|^2}{\omega - \omega_j + i\frac{\gamma}{2}} \right) (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger) \\
 &= -\left(\frac{1}{\omega - \omega_{ph} + i\frac{\kappa}{2}} \right) \Sigma_M (\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger).
 \end{aligned} \tag{C12}$$

where we used the definition of Σ_M in Eq. C5. Collecting the results of Eqs. C8, C10, and C12 into Eq. C7 (with the corresponding superscripts “eff”), we can solve for $\mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger$ to obtain (see Eq. B13),

$$\begin{aligned}
 D^R(\omega) &= \mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger \\
 &= \mathcal{A}\mathcal{G}_0^{\text{eff}} a^\dagger \\
 &= \frac{1}{(\omega - \omega_{ph} + i\frac{\kappa}{2}) - \Sigma_M},
 \end{aligned} \tag{C13}$$

which plugged into Eq. B9 yields Eq. C4.

To conclude, let us compare the calculation in VII C 1 and the current one, which give the same answers for $\langle a(\omega) \rangle$. The latter assumes: (a) weak drive and (b) a stationary initial condition $\rho(t_{\text{in}})$; while the former does not demand these restrictions. However, both assumptions are unnecessary for harmonic systems [74], since their response to a driving field is (a) linear no matter how strong the drive is, (b) independent of initial condition.

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- [1] R. F. Ribeiro, L. A. Martı́nez-Martı́nez, M. Du, J. Campos-Gonzalez-Angulo, and J. Yuen-Zhou, “Polariton chemistry: controlling molecular dynamics with optical cavities,” *Chem. Sci.*, vol. 9, pp. 6325–6339, 2018.
 - [2] R. Chikkaraddy, B. de Nijs, F. Benz, S. J. Barrow, O. A. Scherman, E. Rosta, A. Demetriadou, P. Fox, O. Hess, and J. J. Baumberg, “Single-molecule strong coupling at room temperature in plasmonic nanocavities,” *Nature*, vol. 535, no. 7610, pp. 127–130, 2016.
 - [3] H. Leng, B. Szychowski, M.-C. Daniel, and M. Pelton, “Strong coupling and induced transparency at room temperature with single quantum dots and gap plasmons,” *Nat. Commun.*, vol. 9, no. 1, p. 4012, 2018.
 - [4] O. Bitton, S. N. Gupta, and G. Haran, “Quantum dot plasmonics: from weak to strong coupling,” *Nanophotonics*, vol. 8, no. 4, pp. 559–575, 2019.
 - [5] T. S. Haugland, C. Schäfer, E. Ronca, A. Rubio, and H. Koch, “Intermolecular interactions in optical cavities: An ab initio qed study,” *The Journal of Chemical Physics*, vol. 154, no. 9, 2021.
 - [6] A. Koner, M. Du, S. Pannir-Sivajothi, R. H. Goldsmith, and J. Yuen-Zhou, “A path towards single molecule vibrational strong coupling in a fabry–pérot microcavity,” *Chemical Science*, vol. 14, no. 28, pp. 7753–7761, 2023.
 - [7] J. A. Hutchison, T. Schwartz, C. Genet, E. Devaux, and T. W. Ebbesen, “Modifying chemical landscapes by coupling to vacuum fields,” *Angew. Chem., Int. Ed.*, vol. 51, no. 7, pp. 1592–1596, 2012.

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- [8] A. Thomas, L. Lethuillier-Karl, K. Nagarajan, R. M. A. Vergauwe, J. George, T. Chervy, A. Shalabney, E. Devaux, C. Genet, J. Moran, and T. W. Ebbesen, "Tilting a ground-state reactivity landscape by vibrational strong coupling," *Science*, vol. 363, no. 6427, pp. 615–619, 2019.
- [9] D. M. Coles, N. Somaschi, P. Michetti, C. Clark, P. G. Lagoudakis, P. G. Savvidis, and D. G. Lidzey, "Polariton-mediated energy transfer between organic dyes in a strongly coupled optical microcavity," *Nat. Mater.*, vol. 13, no. 7, pp. 712–719, 2014.
- [10] X. Zhong, T. Chervy, L. Zhang, A. Thomas, J. George, C. Genet, J. A. Hutchison, and T. W. Ebbesen, "Energy transfer between spatially separated entangled molecules," *Angew. Chem., Int. Ed.*, vol. 56, no. 31, pp. 9034–9038, 2017.
- [11] C. A. DelPo, S.-U.-Z. Khan, K. H. Park, B. Kudisch, B. P. Rand, and G. D. Scholes, "Polariton decay in donor-acceptor cavity systems," *The Journal of Physical Chemistry Letters*, vol. 12, no. 40, pp. 9774–9782, 2021.
- [12] S. Kéna-Cohen and S. Forrest, "Room-temperature polariton lasing in an organic single-crystal microcavity," *Nature Photonics*, vol. 4, no. 6, pp. 371–375, 2010.
- [13] J. D. Plumhof, T. Stöferle, L. Mai, U. Scherf, and R. F. Mahrt, "Room-temperature bose-einstein condensation of cavity exciton-polaritons in a polymer," *Nature materials*, vol. 13, no. 3, pp. 247–252, 2014.
- [14] P. Törmä and W. L. Barnes, "Strong coupling between surface plasmon polaritons and emitters: a review," *Rep. Prog. Phys.*, vol. 78, no. 1, p. 013901, 2015.
- [15] Y. Zhu, D. J. Gauthier, S. Morin, Q. Wu, H. Carmichael, and T. Mossberg, "Vacuum rabi splitting as a feature of linear-dispersion theory: Analysis and experimental observations," *Physical Review Letters*, vol. 64, no. 21, p. 2499, 1990.
- [16] M. Schubert, "Polarization-dependent optical parameters of arbitrarily anisotropic homogeneous layered systems," *Physical Review B*, vol. 53, no. 8, p. 4265, 1996.
- [17] A. Yariv and P. Yeh, *Photonics: optical electronics in modern communications*. Oxford university press, 2007.
- [18] F. Wu, D. Finkelstein-Shapiro, M. Wang, I. Rosenkamppf, A. Yartsev, T. Pascher, T. C. Nguyen-Phan, R. Cogdell, K. Börjesson, and T. Pullerits, "Optical cavity-mediated exciton dynamics in photosynthetic light harvesting 2 complexes," *Nature Communications*, vol. 13, p. 6864, Nov 2022.
- [19] B. Xiang, J. Wang, Z. Yang, and W. Xiong, "Nonlinear infrared polaritonic interaction between cavities mediated by molecular vibrations at ultrafast time scale," *Science advances*, vol. 7, no. 19, p. eabf6397, 2021.
- [20] B. S. Simpkins, A. D. Dunkelberger, and I. Vurgaftman, "Control, modulation, and analytical descriptions of vibrational strong coupling," *Chemical Reviews*, vol. 123, no. 8, pp. 5020–5048, 2023.
- [21] A. D. Wright, J. C. Nelson, and M. L. Weichman, "Rovibrational polaritons in gas-phase methane," *Journal of the American Chemical Society*, vol. 145, no. 10, pp. 5982–5987, 2023.
- [22] S. Gunasekaran, R. F. Pinard, and A. J. Musser, "Continuum model of strong light-matter coupling for molecular polaritons," *arXiv preprint arXiv:2308.08744*, 2023.
- [23] C. W. Gardiner and M. J. Collett, "Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation," *Phys. Rev. A*, vol. 31, no. 6, pp. 3761–3774, 1985.
- [24] C. Ciuti and I. Carusotto, "Input-output theory of cavities in the ultrastrong coupling regime: The case of time-independent cavity parameters," *Phys. Rev. A*, vol. 74, no. 3, p. 033811, 2006.
- [25] D. A. Steck, *Quantum and atom optics*, vol. 47. Eugene, Oregon: Oregon Center for Optics and Department of Physics, University of Oregon, 2007.
- [26] S. Portolan, O. Di Stefano, S. Savasta, F. Rossi, and R. Girlanda, "Nonequilibrium langevin approach to quantum optics in semiconductor microcavities," *Physical Review B*, vol. 77, no. 3, p. 035433, 2008.
- [27] H. Li, A. Piryatinski, J. Jerke, A. R. S. Kandada, C. Silva, and E. R. Bittner, "Probing dynamical symmetry breaking using quantum-entangled photons," *Quantum Sci. Technol.*, vol. 3, no. 1, p. 015003, 2017.
- [28] R. F. Ribeiro, A. D. Dunkelberger, B. Xiang, W. Xiong, B. S. Simpkins, J. C. Owrutsky, and J. Yuen-Zhou, "Theory for nonlinear spectroscopy of vibrational polaritons," *J. Phys. Chem. Lett.*, vol. 9, no. 13, pp. 3766–3771, 2018.
- [29] M. Reitz, C. Sommer, and C. Genes, "Langevin approach to quantum optics with molecules," *Physical review letters*, vol. 122, no. 20, p. 203602, 2019.
- [30] K. S. Kansanen, J. J. Toppari, and T. T. Heikkilä, "Polariton response in the presence of brownian dissipation from molecular vibrations," *The Journal of Chemical Physics*, vol. 154, no. 4, 2021.
- [31] T. Leppälä, A. G. Abdelmagid, H. A. Qureshi, K. S. Daskalakis, and K. Luoma, "Linear optical properties of organic microcavity polaritons with non-markovian quantum state diffusion," 2023.
- [32] E. J. Heller, "The semiclassical way to molecular spectroscopy," *Accounts of Chemical Research*, vol. 14, no. 12, pp. 368–375, 1981.
- [33] S. Mukamel, *Principles of nonlinear optical spectroscopy*. Oxford University Press, 1995.
- [34] D. Tannor, *Introduction to Quantum Mechanics*. University Science Books, 2007.
- [35] E. J. Heller, *The semiclassical way to dynamics and spectroscopy*. Princeton University Press, 2018.
- [36] J. A. Ćwik, P. Kirton, S. De Liberato, and J. Keeling, "Excitonic spectral features in strongly coupled organic polaritons," *Phys. Rev. A*, vol. 93, no. 3, p. 033840, 2016.
- [37] A. Z. Lieberherr, S. T. Furniss, J. E. Lawrence, and D. E. Manolopoulos, "Vibrational strong coupling in liquid water from cavity molecular dynamics," *arXiv preprint arXiv:2305.02711*, 2023.
- [38] M. A. Zeb, P. G. Kirton, and J. Keeling, "Exact states and spectra of vibrationally dressed polaritons," *ACS Photonics*, vol. 5, no. 1, pp. 249–257, 2018.
- [39] N. Makri, "The linear response approximation and its lowest order corrections: An influence functional approach," *The Journal of Physical Chemistry B*, vol. 103, no. 15, pp. 2823–2829, 1999.
- [40] More precisely, ρ_{mol} cannot correspond to a system with population inversion, as a harmonic oscillator cannot reproduce $C_2(t)$ in that case. Hence, this mapping only works if $\beta^{eff}(\omega) \geq 0$.
- [41] A. Nitzan, *Chemical Dynamics in Condensed Phases: Relaxation, Transfer and Reactions in Condensed Molecular Systems*. OUP Oxford, 2006.
- [42] In the special case that all the N molecules are identical,

$$\chi(\omega) = -\lim_{\gamma \rightarrow 0^+} \frac{1}{\hbar} \sum_{a,b} [p_a - p_b] \frac{N \frac{\omega_{ph}}{2\epsilon_0 V} |\langle a|\mu|b \rangle|^2}{\omega - \omega_{ba} + i\frac{\gamma}{2}},$$

where a, b label single-molecule eigenstates, and we used $|\chi|^2 = \frac{\omega_{ph}}{2\hbar\epsilon_0\mathcal{V}}$, where $\mathcal{V}$ is the cavity mode volume. Thus $\chi(\omega)$ is proportional to molecular concentration $\rho_0 = \frac{N}{\mathcal{V}}$. This expression coincides with the textbook expression in [33] (see Eqs. 6.5a and 6.8b) except for a convenient factor of $\frac{\omega_{ph}}{2\epsilon_0}$ that simplifies the notation for our calculations for molecular microcavities.

- [43] Under moderate collective light-matter interaction couplings, if hereafter Eq. 10 were to be used instead of Eq. 28, the errors would be negligible given how off-resonant the non-RWA terms are.
- [44] See for example, Eqs. 2.91 and 2.92 in [33] or Eq. 9.16 in [41].
- [45] J. P. Long and B. S. Simpkins, “Coherent coupling between a molecular vibration and fabry-perot optical cavity to give hybridized states in the strong coupling limit,” *ACS Photonics*, vol. 2, no. 1, pp. 130–136, 2015.
- [46] A. Shalabney, J. George, J. Hutchison, G. Pupillo, C. Genet, and T. W. Ebbesen, “Coherent coupling of molecular resonators with a microcavity mode,” *Nat. Commun.*, vol. 6, p. 6, 2015.
- [47] V. M. Agranovich, M. Litinskaia, and D. G. Lidzey, “Cavity polaritons in microcavities containing disordered organic semiconductors,” *Phys. Rev. B*, vol. 67, no. 8, p. 085311, 2003.
- [48] N. S. Stoyanov, D. W. Ward, T. Feurer, and K. A. Nelson, “Terahertz polariton propagation in patterned materials,” *Nature materials*, vol. 1, no. 2, pp. 95–98, 2002.
- [49] R. Damari, O. Weinberg, D. Krotkov, N. Demina, K. Akulov, A. Golombek, T. Schwartz, and S. Fleischer, “Strong coupling of collective intermolecular vibrations in organic materials at terahertz frequencies,” *Nature Communications*, vol. 10, no. 1, p. 3248, 2019.
- [50] J. B. Pérez-Sánchez, A. Koner, N. P. Stern, and J. Yuen-Zhou, “Simulating molecular polaritons in the collective regime using few-molecule models,” *Proceedings of the National Academy of Sciences*, vol. 120, no. 15, p. e2219223120, 2023.
- [51] M. Tavis and F. W. Cummings, “Exact solution for an n-molecule radiation-field hamiltonian,” *Physical Review*, vol. 170, no. 2, p. 379, 1968.
- [52] H. Deng, G. Weihs, C. Santori, J. Bloch, and Y. Yamamoto, “Condensation of semiconductor microcavity exciton polaritons,” *Science*, vol. 298, no. 5591, pp. 199–202, 2002.
- [53] A. D. Dunkelberger, B. T. Spann, K. P. Fears, B. S. Simpkins, and J. C. Owrutsky, “Modified relaxation dynamics and coherent energy exchange in coupled vibration-cavity polaritons,” *Nat. Commun.*, vol. 7, p. 13504, 2016.
- [54] B. Xiang, R. F. Ribeiro, Y. Li, A. D. Dunkelberger, B. B. Simpkins, J. Yuen-Zhou, and W. Xiong, “Manipulating optical nonlinearities of molecular polaritons by delocalization,” *Science advances*, vol. 5, no. 9, p. eaax5196, 2019.
- [55] A. D. Dunkelberger, A. B. Grafton, I. Vurgaftman, O. O. Soykal, T. L. Reinecke, R. B. Davidson, B. S. Simpkins, and J. C. Owrutsky, “Saturable absorption in solution-phase and cavity-coupled tungsten hexacarbonyl,” *ACS Photonics*, vol. 6, no. 11, pp. 2719–2725, 2019.
- [56] R. Houdré, R. Stanley, and M. Illegems, “Vacuum-field rabi splitting in the presence of inhomogeneous broadening: Resolution of a homogeneous linewidth in an inhomogeneously broadened system,” *Physical Review A*, vol. 53, no. 4, p. 2711, 1996.
- [57] T. Botzung, D. Hagenmüller, S. Schütz, J. Dubail, G. Pupillo, and J. Schachenmayer, “Dark state semilocalization of quantum emitters in a cavity,” *Phys. Rev. B*, vol. 102, no. 14, p. 144202, 2020.
- [58] C. Sommer, M. Reitz, F. Mineo, and C. Genes, “Molecular polaritonics in dense mesoscopic disordered ensembles,” *Physical Review Research*, vol. 3, no. 3, p. 033141, 2021.
- [59] M. Du and J. Yuen-Zhou, “Catalysis by dark states in vibropolaritonic chemistry,” *Physical Review Letters*, vol. 128, no. 9, p. 096001, 2022.
- [60] G. Engelhardt and J. Cao, “Unusual dynamical properties of disordered polaritons in microcavities,” *Physical Review B*, vol. 105, no. 6, p. 064205, 2022.
- [61] T. Gera and K. Sebastian, “Effects of disorder on polaritonic and dark states in a cavity using the disordered tavis-cummings model,” *The Journal of Chemical Physics*, vol. 156, no. 19, p. 194304, 2022.
- [62] T. Gera and K. Sebastian, “Exact results for the tavis-cummings and hulckel hamiltonians with diagonal disorder,” *The Journal of Physical Chemistry A*, vol. 126, no. 32, pp. 5449–5457, 2022.
- [63] J. Dubail, T. Botzung, J. Schachenmayer, G. Pupillo, and D. Hagenmüller, “Large random arrowhead matrices: Multifractality, semilocalization, and protected transport in disordered quantum spins coupled to a cavity,” *Physical Review A*, vol. 105, no. 2, p. 023714, 2022.
- [64] B. Cohn, S. Sufrin, A. Basu, and L. Chuntonov, “Vibrational polaritons in disordered molecular ensembles,” *The Journal of Physical Chemistry Letters*, vol. 13, no. 35, pp. 8369–8375, 2022.
- [65] H.-T. Chen, Z. Zhou, M. Sukharev, J. E. Subotnik, and A. Nitzan, “Interplay between disorder and collective coherent response: Superradiance and spectral motional narrowing in the time domain,” *Physical Review A*, vol. 106, no. 5, p. 053703, 2022.
- [66] E. Gull, A. J. Millis, A. I. Lichtenstein, A. N. Rubtsov, M. Troyer, and P. Werner, “Continuous-time monte carlo methods for quantum impurity models,” *Reviews of Modern Physics*, vol. 83, no. 2, p. 349, 2011.
- [67] D. Segal, A. J. Millis, and D. R. Reichman, “Nonequilibrium transport in quantum impurity models: exact path integral simulations,” *Physical Chemistry Chemical Physics*, vol. 13, no. 32, pp. 14378–14386, 2011.
- [68] G. Cohen and E. Rabani, “Memory effects in nonequilibrium quantum impurity models,” *Physical Review B*, vol. 84, no. 7, p. 075150, 2011.
- [69] E. Gull, D. R. Reichman, and A. J. Millis, “Numerically exact long-time behavior of nonequilibrium quantum impurity models,” *Physical Review B*, vol. 84, no. 8, p. 085134, 2011.
- [70] C.-Y. Hsieh and J. Cao, “A unified stochastic formulation of dissipative quantum dynamics. ii. beyond linear response of spin baths,” *The Journal of chemical physics*, vol. 148, no. 1, 2018.
- [71] L. P. Lindoy and D. E. Manolopoulos, “Simple and accurate method for central spin problems,” *Physical review letters*, vol. 120, no. 22, p. 220604, 2018.
- [72] P. Fowler-Wright, B. W. Lovett, and J. Keeling, “Efficient many-body non-markovian dynamics of organic polaritons,” *Physical Review Letters*, vol. 129, no. 17, p. 173001, 2022.
- [73] J. B. Pérez-Sánchez, F. Mellini, N. C. Giebink, and

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- [74] S. Mukamel and Y. Nagata, "Quantum field, interfer-

ence, and entanglement effects in nonlinear optical spectroscopy," *Procedia Chemistry*, vol. 3, no. 1, pp. 132–151, 2011.